

WFLOW-05: PagerDuty & ServiceNow Integration

Series: WFLOW — Workflows and Alert Notifications | **Notebook:** 5 of 10 |
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Incident Management Automation

Integrate Dynatrace workflows with enterprise incident management platforms. This notebook covers PagerDuty and ServiceNow integration patterns, bi-directional sync, and incident lifecycle management.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Platform subscription

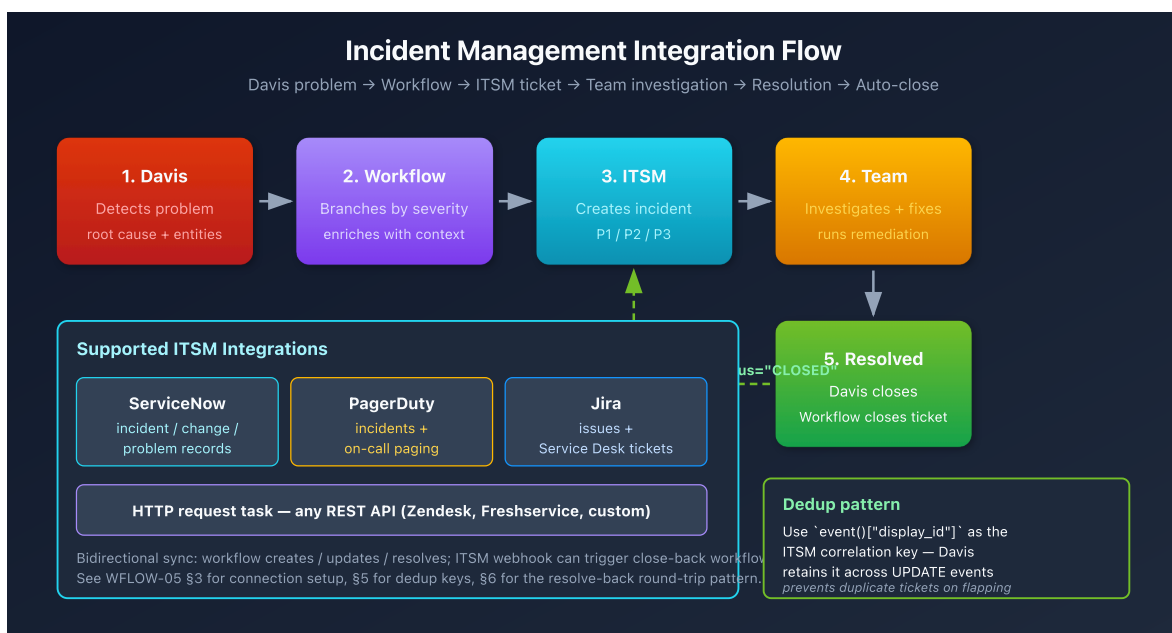
Requirement	Details
Permissions	<code>automation:workflows:write</code> , <code>automation:connections:write</code>
PagerDuty	Admin access to create services and integrations
ServiceNow	Admin access or integration user credentials

1. Integration Overview

Why Integrate?

Benefit	Description
Single pane of glass	Incidents tracked in one system
On-call management	Leverage PagerDuty/ServiceNow schedules
Audit trail	Complete incident history
Escalation	Built-in escalation policies
Metrics	MTTR, incident frequency reporting

Integration Flow



2. PagerDuty Setup

Creating PagerDuty Integration

1. In PagerDuty:

- Go to **Services** → Select or create service
- **Integrations** tab → **Add Integration**
- Select **Events API v2**
- Copy the **Integration Key** (routing key)

2. In Dynatrace:

- Go to **Settings** → **Integration** → **Connections**
- Click **+ Connection**
- Select **PagerDuty**
- Enter the Integration Key
- Name: `pagerduty-production`

Multiple Services Pattern

Create separate connections for different routing:

Connection Name	PagerDuty Service	Use
<code>pagerduty-prod-critical</code>	Production-Critical	P1 incidents
<code>pagerduty-prod-standard</code>	Production-Standard	P2-P4 incidents
<code>pagerduty-platform</code>	Platform-Infrastructure	Infra issues

3. PagerDuty Workflow Tasks

Create Incident

```
name: create_pagerduty_incident
type: dynatrace.pagerduty:create-incident
input:
  connection: pagerduty-production
  severity: '{{ {"CRITICAL": "critical", "HIGH": "error", "MEDIUM":
"warning", "LOW": "info"}.get(event()["severity"], "warning") }}'
  summary: "[{{ event()['severity'] }}] {{ event()['title'] }}"
  source: "dynatrace"
  component: "{{ event().get('root_cause_entity_id', 'unknown') }}"
  group: "{{ event().get('management_zones', ['default'])[0] }}"
  class: "{{ event().get('event_type', 'problem') }}"
  customDetails:
    problem_id: "{{ event()['display_id'] }}"
    problem_url: "{{ event()['problem_url'] }}"
    affected_entities: "{{ event()['affected_entity_ids'] | join(', ') }}"
    root_cause: "{{ event().get('root_cause_entity_id', 'N/A') }}"
    start_time: "{{ event()['start_time'] }}"
  dedupKey: "dynatrace-{{ event()['display_id'] }}"
```

Resolve Incident

Triggered when problem closes:

```
name: resolve_pagerduty_incident
type: dynatrace.pagerduty:resolve-incident
input:
  connection: pagerduty-production
  dedupKey: "dynatrace-{{ event()['display_id'] }}"
  description: "Problem resolved in Dynatrace at {{ event().get('end_time',
now()) }}"
```

Acknowledge Incident

```
name: acknowledge_pagerduty
type: dynatrace.pagerduty:acknowledge-incident
input:
  connection: pagerduty-production
  dedupKey: "dynatrace-{{ event()['display_id'] }}"
```

4. ServiceNow Setup

Creating ServiceNow Connection

1. In ServiceNow:

- Create integration user or use OAuth app
- Grant appropriate roles: `itil`, `rest_api_user`
- Note instance URL: `https://your-instance.service-now.com`

2. In Dynatrace:

- Go to **Settings** → **Integration** → **Connections**
- Click **+ Connection**
- Select **ServiceNow**
- Enter:
 - o Instance URL
 - o Username/Password or OAuth credentials
 - o Name: `servicenow-production`

Authentication Options

Method	Best For
Basic Auth	Quick setup, development
OAuth 2.0	Production, better security
API Key	Service accounts

5. ServiceNow Workflow Tasks

Create Incident

```
name: create_snow_incident
type: dynatrace.servicenow:create-incident
input:
  connection: servicenow-production
  short_description: "[Dynatrace] {{ event()['title'] }}"
  description: |
    A problem has been detected by Dynatrace Dynatrace Intelligence.

    Problem Details:
    =====
    Problem ID: {{ event()['display_id'] }}
    Severity: {{ event()['severity'] }}
    Status: {{ event()['status'] }}
    Start Time: {{ event()['start_time'] }}

    Affected Entities:
    {{ event()['affected_entity_ids'] | join('\n') }}

    Root Cause Entity:
    {{ event().get('root_cause_entity_id', 'Pending analysis') }}

    View in Dynatrace:
    {{ event()['problem_url'] }}
  impact: '{{ {"CRITICAL": 1, "HIGH": 2, "MEDIUM": 2, "LOW": 3}.get(event()
["severity"], 3) }}'
  urgency: '{{ {"CRITICAL": 1, "HIGH": 2, "MEDIUM": 2, "LOW": 3}.get(event()
["severity"], 3) }}'
  category: "Software"
  subcategory: "Application"
  assignment_group: "Platform Engineering"
  caller_id: "dynatrace.integration"
  correlation_id: "DT-{{ event()['display_id'] }}"
```

Update Incident

```
name: update_snow_incident
type: dynatrace.servicenow:update-incident
input:
  connection: servicenow-production
  correlation_id: "DT-{{ event()['display_id'] }}"
  work_notes: |
    [Automated Update from Dynatrace]
    Problem updated at {{ now() }}
    Current status: {{ event()['status'] }}
```

Resolve Incident

```
name: resolve_snow_incident
type: dynatrace.servicenow:update-incident
input:
  connection: servicenow-production
  correlation_id: "DT-{{ event()['display_id'] }}"
  state: 6 # Resolved
  close_code: "Resolved"
  close_notes: |
    Problem automatically resolved in Dynatrace.
    Resolution time: {{ event().get('end_time', now()) }}
```

6. Bi-Directional Sync

Sync Status from ServiceNow to Dynatrace

Use scheduled workflow to check incident status:

```

import { problemsClient } from '@dynatrace-sdk/client-classic-environment-
v2';

export default async function({ connection }) {
  // Query ServiceNow for Dynatrace-related incidents
  const response = await fetch(
    `${connection.instance_url}/api/now/table/incident?` +
    `sysparm_query=correlation_id STARTSWITH DT-^state!=6`,
    {
      headers: {
        'Authorization': `Basic ${btoa(connection.username + ':' +
connection.password)}`,
        'Accept': 'application/json'
      }
    }
  );

  const incidents = await response.json();

  // Process each incident
  for (const incident of incidents.result) {
    const problemId = incident.correlation_id.replace('DT-', '');
    // Add comment to Dynatrace problem with SNOW status
    // ...
  }

  return { processed: incidents.result.length };
}

```

Store Incident ID for Updates

Capture ServiceNow incident number for later updates:


```
tasks:
  - name: create_incident
    type: dynatrace.servicenow:create-incident
    # ... configuration ...

  - name: store_incident_id
    type: dynatrace.automations:run-javascript
    dependsOn: [create_incident]
    input:
      script: |
        export default async function({ result }) {
          const incidentNumber = result('create_incident').sys_id;
          // Store in workflow state or external system
          return { incident_number: incidentNumber };
        }
      }
```

7. Incident Lifecycle Patterns

Complete Lifecycle Workflow

Handle open, update, and close events:

```

name: incident-lifecycle-management

trigger:
  type: davis-problem
  config:
    # Trigger on all problem events (open, update, close)
    entityTagsMatch: all
    entityTags:
      - key: env
        value: prod

conditions:
  - name: is_problem_open
    expression: '{{ event()["status"] == "OPEN" }}'
  - name: is_problem_closed
    expression: '{{ event()["status"] == "CLOSED" }}'

tasks:
  # CREATE: When problem opens
  - name: create_incident
    type: dynatrace.pagerduty:create-incident
    conditions: [is_problem_open]
    input:
      connection: pagerduty-production
      severity: critical
      summary: "{{ event()['title'] }}"
      dedupKey: "dynatrace-{{ event()['display_id'] }}"

  # RESOLVE: When problem closes
  - name: resolve_incident
    type: dynatrace.pagerduty:resolve-incident
    conditions: [is_problem_closed]
    input:
      connection: pagerduty-production
      dedupKey: "dynatrace-{{ event()['display_id'] }}"

```

Deduplication Strategy

Platform	Dedup Field	Pattern
PagerDuty	dedupKey	dynatrace-{problem_id}
ServiceNow	correlation_id	DT-{problem_id}

Same problem ID = same incident (no duplicates).

Monitor Incident Integration

```
// PagerDuty/ServiceNow task executions
fetch events, from: now() - 7d
| filter event.type == "automation.task.execution"
| filter contains(task.type, "pagerduty") or contains(task.type,
"servicenow")
| summarize
    total = count(),
    succeeded = countIf(task.status == "SUCCEEDED"),
    failed = countIf(task.status == "FAILED"),
    by:{task.type}
| fieldsAdd success_rate = round(100.0 * succeeded / total, decimals: 2)
| sort total desc
```

```
// Failed incident management tasks
fetch events, from: now() - 24h
| filter event.type == "automation.task.execution"
| filter task.status == "FAILED"
| filter contains(task.type, "pagerduty") or contains(task.type,
"servicenow")
| fields timestamp, workflow.name, task.name, task.type, task.error
| sort timestamp desc
| limit 20
```

```
// Incidents created per day
fetch events, from: now() - 30d
| filter event.type == "automation.task.execution"
| filter contains(task.type, "pagerduty:create") or contains(task.type,
"servicenow:create")
| filter task.status == "SUCCEEDED"
| summarize incidents_created = count(), by:{time_bucket = bin(timestamp,
24h)}
| sort time_bucket asc
```

Next Steps

With incident management configured, customize message templates:

Recommended Path

1. **WFLOW-06: Custom Notification Templates** - Rich formatting
2. **WFLOW-07: Problem-Triggered Remediation** - Auto-remediation
3. **WFLOW-08: JavaScript & HTTP Actions** - Custom integrations

Key Takeaways

- **PagerDuty** uses Events API v2 with routing keys
 - **ServiceNow** uses REST API with correlation IDs
 - **Deduplication** prevents duplicate incidents
 - **Lifecycle workflows** handle open/update/close events
 - **Bi-directional sync** keeps systems in alignment
-

Summary

In this notebook, you learned:

- Integration benefits and architecture
 - PagerDuty connection setup and tasks
 - ServiceNow connection setup and tasks
 - Bi-directional sync patterns
 - Incident lifecycle management
 - Deduplication strategies
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References

- [PagerDuty action \(DT docs\)](#)
- [ServiceNow action \(DT docs\)](#)
- [Notification actions umbrella \(DT docs\)](#)
- [HTTP request action \(DT docs\)](#)
- [Davis Problems app \(DT docs\)](#)

- [PagerDuty Events API v2 \(PagerDuty Developer\)](#)
- [ServiceNow REST API reference \(ServiceNow Developer\)](#)

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